

Fasted, average C labeling

13C tracer infused

Blood

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	14.7	9.4	6.6	1.6	0	0	0		1.1	0.1
lactate	5.4	12.1	12.2	4.1	0.1	0	0		3.6	0.2
alanine	3.6	4.2	17.3	1.4	0	0	0			0
glutamine	1.3	1.1	0.7	19.7	0	0	0		1.4	0
palmitate	0.3	0.3	0.3	0.7	12.3	0.5	0		0.2	1.5
oleate	0.3	0.4	0.5	0.5	0	5.7	0		0.3	1.3
linoleate	0.5	0.7	0.3	1	0	0	7.9		0.3	2.4
glycerol	4.2	3	2.2	0.5	0	0.1	0	12.2	0.1	0
citrate	1	0.7	0.2	0.7	0.1	0	0		30.9	0
3-HB	0.2	0.2	0	0.4	0	0	0		0.3	8.6

labeled metabolites

Lv

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	2.1	2.3	2.3	0.4	1.2					
lactate	4	5.3	5.5	2.9	3.9					
alanine	1.2	3	3.1	1.9	1.8					
glutamine	2.1	2.6	3.2		4.6					
palmitate	0.3	0.6	0.7	0.1	0.7					
oleate	0.3	0.6	0.6		0.7					
linoleate	0.5	1	1.3		1.2					
glycerol	0.3	0.9	0.8	0.2	0.4					
citrate	13.1	0.3	0.3	0.5	0.3					
3-HB	0.2	0.3	0.3		0.4					

S

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	1.2	1.7	1.5	0.8	1.4					
lactate	3	3.9	3.8	3.6	3.7					
alanine	0.7	1	0.9	0.9	0.8					
glutamine	2.6	5.4	5.4		5.8					
palmitate	0.3	1.3	1.3	0.6	1.3					
oleate	0.4	1	1		1					
linoleate	0.8	2.1	2.2		2.2					
glycerol	0.2	0.5	0.5	0.1	0.5					
citrate	6.4	0.3	0.2	0.4	0.3					
3-HB	0.5	1	1.1		1.1					

citrate
malate
succinate
a-ketoglutarate
glutamate

M

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	4	3.2	3.2	1.2	2.6					
lactate	4	5.1	5.1	4.6	4.6					
alanine	0.7	0.8	0.8	0.7	0.8					
glutamine	0.7	1.6	1.6		2.2					
palmitate	0	1	0.9	1	0.8					
oleate	0.4	1.1	1		1					
linoleate	0.5	1.7	1.7		1.4					
glycerol	0.3	0.5	0.4	0.1	0.4					
citrate	14.4	0.2	0.1	0.3	0.2					
3-HB	0.1	0.4	0.3		0.4					

I

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	1.5	1.7	1.6	1	1.7					
lactate	4.1	5.4	5	4.6	5.5					
alanine	0.6	1.4	1.4	1.3	1.2					
glutamine	2.5	5.3	6.3		5.6					
palmitate	0.1	0.9	1	0.1	1					
oleate	0.3	0.7	0.7		0.8					
linoleate	0.7	1.7	1.6		1.7					
glycerol	0.1	0.5	0.6	0.1	0.4					
citrate	9.3	0.4	0.3	0.4	0.4					
3-HB	0.8	1.2	1.3		1.4					

citrate
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glutamate

H

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	1.7	2	2.1	0.1	2					
lactate	3	3.8	3.8		3.7					
alanine	0.3	0.4	0.4	0.3	0.3					
glutamine	0.8	0.5	0.3		0.6					
palmitate	0.2	1.8	2	1.1	1.9					
oleate	0.5	1.2	1.4		1.3					
linoleate	0.8	1.9	2.1		2					
glycerol	0.2	0.3	0.2	0.1	0.3					
citrate	15	0.1	0.1	0.2	0.1					
3-HB	0.4	0.6	0.5		0.6					

Br

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	9.2	11	11.2	12.2	10.9					
lactate	5.3	6.8	6.8	6.6	6.8					
alanine										
glutamine	0.4	1.2	1.2		1.2					
palmitate	0.1	0.3	0.3	0.2	0.4					
oleate	0.3	0.4	0.3		0.4					
linoleate	0.5	0.8	0.7		0.7					
glycerol	2	3.4	3.4	3.2	2.8					
citrate	8.2	0.7	0.7	0.8	0.7					
3-HB	0.3	0.4	0.3		0.4					

citrate
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glutamate

Ln

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	1.7	2.9	2.2	2	1.9					
lactate	3.6	5.6	4.4	4.4	3.7					
alanine	0.7	2.1	1.4	1.4	1					
glutamine	1	2	1.5		2.3					
palmitate	0.3	1.4	1.5	0.5	1.7					
oleate	0.4	0.9	0.9		1					
linoleate	0.6	1.3	1.7		1.3					
glycerol	0.2	0.8	0.6	0.5	0.6					
citrate	10.8	0.5	0.3	0.6	0.3					
3-HB	0.2	0.5	0.5		0.5					

Bat

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	1.9	2.7	2.6	0.7	1.7					
lactate	3	4	3.7	5	2.8					
alanine	0.7	1.3	1.4	0.8	0.8					
glutamine	0.9	1.5	1.5		3.3					
palmitate		0.7	0.7	0.2	0.8					
oleate										
linoleate										
glycerol										
citrate	0.5	0.6	0.6	0.3	0.4					
3-HB	18.1	0.6	0.3	1.1	0.4					

citrate
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succinate
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glutamate

K

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	1.8	2.7	2.7	0.6	2					
lactate	4.4	6.5	6.6	5.7	5.2					
alanine	0.9	2.2	1.9	1.2	1.7					
glutamine	1.3	2.5	2.9		3.2					
palmitate	0.2	1	1.1	0.7	1.1					
oleate	0.4	0.9	1		0.9					
linoleate	1.3	2.2	3.1		2					
glycerol	0.4	1.2	1.1	0.5	0.6					
citrate	41.4	4.1	5	5.6	5.2					
3-HB	0.9	1.1	1.2		1.2					

gW

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	1.4	2.1	2.3	0.9	1.4					
lactate	2.5	3.9	4.3		3					
alanine		2.1	2.8	2.9	2.4					
glutamine	1.1	2.3	2.2		4.1					
palmitate		0.4	0.4	0.3	0.3					
oleate	0.3	0.6	0.7		0.5					
linoleate	0									
glycerol	2.1	3.6	0.4	0	0.2					
citrate		0.2	0.1		0.3					
3-HB	0.2	0.4	0.5		0.5					

citrate
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P

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	0.8	1.9	1.9	1.2	1.7					
lactate	3.5	5	4.9	3.6	4.5					
alanine	1.2	3.2	2.7	2	2					
glutamine	3.4	5.4	5.8		7.5					
palmitate	0.2	0.8	0.9	0.5	0.9					
oleate	0.2	0.7	0.6		0.7					
linoleate	0.4	1.3	1.4		1.3					
glycerol	0.2	0.5	0.5	0.1	0.4					
citrate	2.7	0.3	0.4	0.4	0.5					
3-HB	0.2	0.7	0.7		0.7					

iW

	glucose	lactate	alanine	glutamine	palmitate	oleate	linoleate	glycerol	citrate	3-HB
glucose	1.4	1.9	2.1	1.2	1.3					
lactate	3.3	4	4.7	6.9	3.2					
alanine		1.3	1.9							
glutamine	1.3	2.3	2.3		3.5					
palmitate	0	0.9	0.3		0.7					
oleate										
linoleate										
glycerol										
citrate	29.7	1.7	0.2		0.3					
3-HB										

citrate
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a-ketoglutarate
glutamate

labeled metabolites